

The wave and Schrödinger Gagliardo–Nirenberg conjectures

Candidate full solution to the closing conjecture of arXiv:2201.13050

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Status: candidate full proof, likely valid. Both exponent regions conjectured by Mandel follow by globalizing the paper’s compact-piece theorem with ordinary, respectively parabolic, Littlewood–Paley decompositions.

1 Source conjecture

Let $z = (x, t) \in \mathbb{R}^{d-1} \times \mathbb{R}$. Mandel studies

$$\|u\|_q \lesssim \|\square u\|_r^{1-\kappa} \|u\|_r^\kappa, \quad \square = \partial_{tt} - \Delta_x, \quad (\text{W})$$

$$\|v\|_q \lesssim \|(i\partial_t - \Delta_x)v\|_r^{1-\kappa} \|v\|_r^\kappa. \quad (\text{S})$$

The source is R. Mandel, *On Gagliardo–Nirenberg inequalities with vanishing symbols*, arXiv:2201.13050, Anal. PDE 17 (2024), 3447–3476 [1].

ON GAGLIARDO-NIRENBERG INEQUALITIES WITH VANISHING SYMBOLS

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We conjecture that at least for $1 < r \leq 2 \leq q < \infty$ and $0 < \kappa < 1$ the inequality (27) actually holds for exponents

$$(29) \quad \frac{1}{r} - \frac{1}{q} = \frac{2(1-\kappa)}{d}, \quad \min \left\{ \frac{1}{r}, \frac{1}{q'} \right\} \geq \frac{d-2\kappa}{2(d-1)}$$

whereas the corresponding inequality involving the Schrödinger operator holds whenever

$$\frac{1}{r} - \frac{1}{q} = \frac{2(1-\kappa)}{d+1}, \quad \min \left\{ \frac{1}{r}, \frac{1}{q'} \right\} \geq \frac{d+1-2\kappa}{2d}.$$

Note that the Sobolev inequalities [20, Theorem 1.1] then take the form of the endpoint estimate $\kappa = 0$ in (29).

ACKNOWLEDGMENTS

Figure 1: The two conjectured exponent regions on source PDF page 29.

Precisely, for $1 < r \leq 2 \leq q < \infty$ and $0 < \kappa < 1$, the paper conjectures (W) under

$$\frac{1}{r} - \frac{1}{q} = \frac{2(1-\kappa)}{d}, \quad \min \left\{ \frac{1}{r}, \frac{1}{q'} \right\} \geq \frac{d-2\kappa}{2(d-1)}, \quad (1)$$

and (S) under

$$\frac{1}{r} - \frac{1}{q} = \frac{2(1 - \kappa)}{d + 1}, \quad \min \left\{ \frac{1}{r}, \frac{1}{q'} \right\} \geq \frac{d + 1 - 2\kappa}{2d}. \quad (2)$$

The wave statement is understood for $d \geq 2$, and the Schrödinger statement for $d \geq 2$, as in the source section.

2 Proof intuition

The hard oscillatory-integral work is already local in the source. On one Fourier annulus, the wave cone has $d - 2$ nonzero principal curvatures and the paraboloid has $d - 1$. Substituting these numbers and the vanishing orders $(1, 0)$ into the source's compact-piece theorem gives exactly (1) and (2). Homogeneity makes the constant independent of the annulus.

It remains to add the annuli without losing the multiplicative right-hand side. This is precisely where the conjectural restriction $r \leq 2 \leq q$ matters. Littlewood–Paley theory first places the output annuli in ℓ^2 . Sequence Hölder separates the two factors. Finally, L^r is 2-concave for $r \leq 2$, while L^q is 2-convex for $q \geq 2$; hence both ℓ^2 input sums collapse to the two global norms. The paraboloid uses parabolic rather than isotropic annuli. A direct one-dimensional normal estimate handles the flat $d = 2$ wave cone.

3 Two square-function facts

We use either the ordinary frequency dilations $\xi \mapsto \lambda\xi$ or the parabolic dilations

$$\delta_\lambda(\xi', \tau) = (\lambda\xi', \lambda^2\tau). \quad (3)$$

For the latter, a convenient homogeneous quasi-norm is $\rho(\xi', \tau) = (|\xi'|^4 + \tau^2)^{1/4}$. A smooth partition of unity on $\mathbb{R}^d \setminus \{0\}$ gives annular multipliers $(\Delta_j)_{j \in \mathbb{Z}}$. In both dilation systems the Littlewood–Paley theorem says

$$\|f\|_p \asymp \left\| \left(\sum_j |\Delta_j f|^2 \right)^{1/2} \right\|_p, \quad 1 < p < \infty. \quad (4)$$

For the parabolic system this is the standard anisotropic square-function theorem; it also follows from random signs, the anisotropic Mikhlin theorem, and Khintchine's inequality [2].

We shall additionally use the elementary inequalities

$$\left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_q \leq \left(\sum_j \|f_j\|_q^2 \right)^{1/2}, \quad q \geq 2, \quad (5)$$

$$\left(\sum_j \|f_j\|_r^2 \right)^{1/2} \leq \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_r, \quad 1 \leq r \leq 2. \quad (6)$$

The first is the triangle inequality in $L^{q/2}$. Raising the second to the r th power reduces it to Minkowski's integral inequality in $\ell^{2/r}$.

4 Uniform annular estimates

Lemma 1 (Wave annulus). *Suppose $d \geq 3$, $0 < \kappa < 1$, and $1 < r \leq 2 \leq q < \infty$ satisfy (1). If $\text{supp } \widehat{w} \subset \{\xi : 1/2 \leq |\xi| \leq 2\}$, then*

$$\|w\|_q \leq C \|\square w\|_r^{1-\kappa} \|w\|_r^\kappa. \quad (7)$$

The same constant works on every isotropic dilate of the annulus.

Proof. Away from the characteristic cone, the symbol is bounded away from zero, so Bernstein and a smooth inverse multiplier give (7). Near the cone, use a finite smooth partition into compact graph pieces. The cone away from its vertex is smooth and has $k_0 = d - 2$ nonzero principal curvatures. Apply Theorem 4 and Remark 1(d) of [1] with vanishing orders $\alpha_1 = 1$, $\alpha_2 = 0$. Their averaged order is $\bar{\alpha} = 1 - \kappa$, and the two local conditions become

$$\frac{1}{r} - \frac{1}{q} \geq \frac{2\bar{\alpha}}{k_0 + 2} = \frac{2(1 - \kappa)}{d}, \quad \min \left\{ \frac{1}{r}, \frac{1}{q'} \right\} \geq \frac{k_0 + 2\bar{\alpha}}{2(k_0 + 1)} = \frac{d - 2\kappa}{2(d - 1)}.$$

The non-strict inequalities are allowed because $0 < \bar{\alpha} < 1$, $\alpha_1 \neq \alpha_2$, and $0 < \kappa < 1$. Finite recombination proves the estimate. The wave symbol has isotropic degree two, and $d(1/r - 1/q) = 2(1 - \kappa)$, so dilation gives the same constant on every annulus. $\square$

Lemma 2 (Schrödinger annulus). *Suppose $d \geq 2$ and the exponents satisfy (2). If $\text{supp } \widehat{w} \subset \{1/2 \leq \rho \leq 2\}$, then*

$$\|w\|_q \leq C \|(i\partial_t - \Delta_x)w\|_r^{1-\kappa} \|w\|_r^\kappa. \quad (8)$$

The same constant works on all parabolic dilates.

Proof. The characteristic paraboloid has $k_0 = d - 1$ nonzero principal curvatures. The same substitution in the source theorem gives

$$\frac{2\bar{\alpha}}{k_0 + 2} = \frac{2(1 - \kappa)}{d + 1}, \quad \frac{k_0 + 2\bar{\alpha}}{2(k_0 + 1)} = \frac{d + 1 - 2\kappa}{2d}.$$

Away from the paraboloid use Bernstein and an inverse multiplier. Under (3), the homogeneous dimension is $Q = (d - 1) + 2 = d + 1$ and the Schrödinger symbol has degree two. The identity $Q(1/r - 1/q) = 2(1 - \kappa)$ makes the constant dilation-invariant. $\square$

Lemma 3 (Flat two-dimensional wave annulus). *Lemma 1 also holds when $d = 2$.*

Proof. Near either characteristic line, take normal/tangent Fourier coordinates (η, σ) . On a unit annular patch, $|\sigma|$ is bounded above and below, and the wave symbol is a nonzero constant multiple of $\eta\sigma$. Thus $\|\partial_y w\|_r \asymp \|\square w\|_r$, where y is the normal physical coordinate. Put $a = 1 - \kappa = 1/r - 1/q$. For every tangent coordinate x , the one-dimensional Gagliardo–Nirenberg inequality gives

$$\|w(x, \cdot)\|_{L_y^q} \lesssim \|\partial_y w(x, \cdot)\|_{L_y^r}^a \|w(x, \cdot)\|_{L_y^r}^{1-a}.$$

Hölder in x and vector-valued Bernstein for the compact tangent-frequency support bound the two resulting $L_x^q(L_y^r)$ norms by the corresponding global L^r norms. This proves the local estimate. Away from the lines use the inverse-multiplier argument, and finite recombination finishes the unit annulus. Scale invariance gives all annuli. $\square$

5 Full solution

Theorem 1. *Both conjectures on source PDF page 29 are true. More explicitly:*

1. *If $d \geq 2$ and (1) holds, then (W) holds for every $u \in \mathcal{S}(\mathbb{R}^d)$.*
2. *If $d \geq 2$ and (2) holds, then (S) holds for every $v \in \mathcal{S}(\mathbb{R}^d)$.*

Proof. First take the ordinary annular projections and write $A_j = \|\Delta_j \square u\|_r$, $B_j = \|\Delta_j u\|_r$. Since constant coefficient multipliers commute, Lemmas 1 and 3, (4), and (5) give

$$\|u\|_q \lesssim \left(\sum_j A_j^{2(1-\kappa)} B_j^{2\kappa} \right)^{1/2}.$$

Hölder in the sequence variable, followed by (6) and (4), yields

$$\begin{aligned} \|u\|_q &\lesssim \left(\sum_j A_j^2 \right)^{(1-\kappa)/2} \left(\sum_j B_j^2 \right)^{\kappa/2} \\ &\lesssim \|\square u\|_r^{1-\kappa} \|u\|_r^\kappa. \end{aligned}$$

This is (W).

For (S), repeat the identical computation with the parabolic annular projections and Lemma 2. The anisotropic version of (4) supplies both square-function bounds. This proves the Schrödinger assertion. $\square$

6 Verification, scope, and novelty

The proof uses the source’s own proved compact-piece theorem, plus standard square-function theory. A deterministic check in the packet verifies both curvature substitutions, both homogeneous dimensions, and the sequence inequalities on randomized finite data. The LaTeX packet was rendered and visually inspected page by page.

The theorem covers the full stated ranges, including equality in the curvature thresholds. It does not address $r = 1$, $q = \infty$, operators with general unbounded characteristic sets, optimal constants, or extremizers.

Bounded novelty searches on 13 August 2026 covered the run indexes, the exact title and conjecture language, wave/Schrödinger Gagliardo–Nirenberg phrases, the displayed exponent forms, and later works citing the source. No later primary answer was found. Novelty confidence is moderate: the globalization is short and may be known informally.

Human-review recommendation. Verify that Theorem 4 plus Remark 1(d) of [1] applies uniformly to the finitely many unit-annulus cone and paraboloid patches, and check the anisotropic Littlewood–Paley normalization. The global summation is elementary once those annular estimates are accepted.

References

- [1] R. Mandel, *On Gagliardo–Nirenberg inequalities with vanishing symbols*, Anal. PDE **17** (2024), 3447–3476; <https://arxiv.org/abs/2201.13050>.
- [2] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton Mathematical Series 43, Princeton University Press, 1993.